

Notes: Ho (AER, 2009)

Insurer-Provider Networks in the Medical Care Market

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1 Big picture

- **Question.** How are hospital-insurer networks formed, and what does the observed pattern of inclusions and exclusions reveal about the bargaining power of hospitals and health plans?
- **Setting.** Managed care plans choose hospital networks; consumers choose plans partly based on the hospitals they expect to access; sick consumers then choose hospitals within their plans' networks.
- **Core empirical challenge.** The data contain hospital networks but not the negotiated hospital prices. The paper therefore infers relative bargaining power from the networks that are observed and the networks that are not observed.
- **Main mechanism.** A plan wants attractive hospitals in its network because they increase enrollee demand and allow higher premiums, but attractive hospitals, capacity-constrained hospitals, and hospitals in systems can demand higher payments.
- **Demand input.** The paper uses hospital-demand and plan-demand estimates from Ho (2006) to calculate the producer surplus generated by alternative hospital networks.
- **Supply-side method.** The supply model uses revealed-preference inequalities: the observed network must generate at least as much expected profit as small deviations that add or drop hospitals.
- **Main result.** Star hospitals, capacity-constrained hospitals, and hospitals in systems capture higher markups than other hospitals.
- **Magnitude.** Star hospitals are estimated to capture roughly \$6,700 more per patient than other hospitals. Capacity-constrained hospitals receive similarly large incremental markups when that variable is used instead of the star-hospital measure.
- **Systems.** Hospitals in systems capture a larger fixed surplus component, and plans appear to face a penalty when they contract with some but not all hospitals in the same system.
- **Investment implications.** Hospitals have incentives to become more attractive to consumers, to merge into systems, and possibly to maintain capacity constraints if scarcity improves bargaining leverage.
- **Economic takeaway.** Hospital network formation is not explained by total producer surplus alone. Bargaining power and market structure matter for which contracts are observed and for how surplus is divided.

2 Incremental contribution of the paper

Incremental contribution: Ho's paper makes several contributions relative to prior work on managed care, hospital competition, and vertical contracting.

1. **From demand-only to network formation.** Earlier work measured how consumers value hospitals and insurers. This paper embeds demand estimates into a supply-side model of how insurer-provider networks are chosen.
2. **Inference without price data.** Hospital-insurer negotiated prices are usually confidential. The paper shows that observed networks alone, combined with demand estimates and cost data, can identify bounds on the profits captured by different hospital types.
3. **Large-network empirical method.** The paper applies the Pakes et al. revealed-preference inequality approach to a setting with many buyers, many sellers, and many potential contracts. This is a practical alternative to solving a full two-sided matching or bargaining equilibrium with multiple equilibria.
4. **Bargaining channel in healthcare.** It distinguishes producer surplus generated by a contract from the share of that surplus captured by hospitals. The paper shows that positive surplus is not enough to predict contracting: some positive-surplus contracts are not formed because hospitals may demand too much.

5. **Three sources of hospital bargaining power.** The paper provides evidence that patient appeal, capacity constraints, and system membership all increase provider leverage.
6. **Long-run investment incentives.** By estimating which characteristics raise hospital markups, the paper connects short-run contracting outcomes to hospitals' incentives to invest in quality, technology, capacity, and mergers.
7. **Network design and welfare.** The analysis clarifies why narrow networks may arise even when broader networks would create positive gross surplus, and why network restrictions can distort plan choice and hospital access.

3 Data and measurement

3.1 Industry timing and network decisions

The model organizes the insurer-provider market into five stages:

1. hospitals make price offers to plans;
2. plans choose their hospital networks;
3. plans set premiums;
4. consumers and employers choose plans;
5. sick enrollees visit hospitals and plans pay hospitals per service.

Interpretation: The empirical analysis focuses on stages 1 and 2, but the model must account for stages 3–5 because networks affect premiums, plan demand, hospital demand, and costs.

3.2 Plan and hospital datasets

The paper combines multiple datasets:

- **Atlantic Information Services (AIS):** managed care plans in 43 major U.S. markets in 2002, including premiums, enrollment, plan type, and networks.
- **NCQA/HEDIS and CAHPS:** clinical quality and member-satisfaction measures for plans.
- **American Hospital Association (AHA):** hospital characteristics such as beds, teaching status, ownership, nurses, doctors, services, and capacity.
- **MEDSTAT MarketScan:** privately insured inpatient claims used to estimate hospital choice.

Interpretation: The paper deliberately combines plan-level, hospital-level, and encounter-level data. Demand is estimated from patient-level utilization, while supply-side bargaining is inferred from plan-hospital network links.

3.3 Analysis sample

The demand-side dataset contains 665 hospitals and 516 managed care plans. For the supply-side network analysis, the paper excludes Baltimore, Kaiser plans, plans with unobserved premiums, and a small number of outlier selective plans. The remaining supply-side sample contains 441 plans in 42 markets and 633 non-Kaiser hospitals.

Interpretation: The supply-side sample is restricted to markets where hospital prices are not centrally regulated and where plans make meaningful network choices.

3.4 Hospital characteristics

The paper focuses on hospital characteristics that may affect consumer demand and bargaining power:

- number of beds;
- teaching status;
- ownership status;
- nurses per bed;
- service indices for cardiac, imaging, cancer, and birth services;
- whether the hospital is a star hospital;
- whether the hospital is predicted to be capacity constrained;
- whether the hospital belongs to a hospital system.

Interpretation: Some characteristics are directly observed; star status and predicted capacity constraints are constructed using the estimated demand model. These variables are meant to capture bargaining-relevant scarcity and consumer appeal, not merely clinical quality.

3.5 Plan characteristics

Plan controls include:

- premiums per member per month;
- physician contracts per 1,000 population;
- plan clinical quality measures;
- member satisfaction measures;
- plan age;
- carrier fixed effects for major carriers;
- plan type, especially POS versus HMO.

Interpretation: Plan demand depends on both price and nonprice quality. The hospital network enters plan utility through the expected utility that consumers receive from access to hospitals.

3.6 Network selectivity

A plan is more selective when it excludes more major hospitals. Markets are classified by the extent to which large plans exclude large hospitals. Figure 1 shows that many plans include all major hospitals, but a substantial number exclude one or more.

Interpretation: The variation in network selectivity is the raw empirical variation used to learn about bargaining. If all plans included all hospitals, there would be little information about which hospitals can profitably demand high prices.

4 Models and empirical specifications

4.1 Hospital demand

Consumer i of diagnosis l receives utility from hospital h :

$$u_{i,h,l} = \eta_h + x_h\alpha + x_h v_{i,l}\beta + \varepsilon_{i,h,l}. \quad (1)$$

Here x_h are observed hospital characteristics, η_h are unobserved hospital characteristics, and $v_{i,l}$ are patient characteristics such as diagnosis and location.

Interpretation: Hospital demand converts hospital characteristics into patient flows. It is crucial because a hospital's bargaining leverage depends on how much consumers value access to that hospital.

4.2 Expected utility from a hospital network

Expected utility from the hospital network of plan j in market m is:

$$EU_{i,j,m} = \sum_l p_{i,l} \log \left[\sum_{h \in H_j} \exp(\eta_h + x_h\alpha + x_h v_{i,l}\beta) \right], \quad (2)$$

where H_j is the hospital network of plan j .

Interpretation: This formula summarizes the value of network breadth and network quality. It is not just the number of hospitals; it is the expected log-sum value of access to hospitals that are attractive for the consumer's likely illnesses.

4.3 Plan demand

Consumer i 's utility from plan j in market m is:

$$u_{i,j,m} = \xi_{j,m} + z_{j,m}\lambda + \gamma_1 EU_{i,j,m} + \gamma_2 \frac{\text{premium}_{j,m}}{y_i} + \omega_{i,j,m}. \quad (3)$$

Interpretation: The expected hospital-network utility enters plan demand directly. This is how a hospital can create value for a plan: better access attracts enrollees and supports higher premiums.

4.4 Producer surplus from a network

The producer surplus generated by plan j with hospital network H_j is:

$$S_{j,m}(H_j, H_{-j}) = \sum_i n_i \left[s_{i,j,m}(H_j, H_{-j}) \text{premium}_{j,m} - p_i \sum_{h \in H_j} s_{i,h}(H_j) \text{cost}_h \right]. \quad (4)$$

Interpretation: Producer surplus is the gross profit generated by a plan's hospital network before splitting that profit between the plan and the hospitals. A contract can generate positive producer surplus but still fail if the hospital demands too much of the surplus.

4.5 Plan profit

Plan profits equal producer surplus minus hospital payments and nonhospital costs:

$$\pi_{j,m}^P(H_j, H_{-j}, X, \theta) = S_{j,m}(H_j, H_{-j}) - C_{j,m}^{HOSP}(H_j, H_{-j}, X, \theta) - C_{j,m}^{NHOSP}(H_j, H_{-j}, X, \theta). \quad (5)$$

Interpretation: The object estimated in the supply model is not just surplus, but surplus net of the hospital profits implied by bargaining.

4.6 Revealed-preference inequalities

The core plan-side inequality is:

$$E[\pi_{j,m}^P(H_j, H_{-j}, X, \theta) \mid I_{j,m}] \geq E[\pi_{j,m}^P(H_j^h, H_{-j}, X, \theta) \mid I_{j,m}], \quad (6)$$

where H_j^h is the alternative network formed by reversing plan j 's contract with hospital h .

Interpretation: If plan j accepted or rejected a contract in the observed data, the model assumes this choice was at least as profitable in expectation as a small deviation. These inequalities identify a set of parameters consistent with observed choices.

4.7 Hospital profit components

The hospital profit function includes:

- a fixed component for system membership;
- a penalty for dropping a same-system hospital;
- a per-patient component for star hospitals;
- a per-patient component for capacity-constrained hospitals;
- a cost per admission term.

Interpretation: These components ask which observable hospital characteristics allow hospitals to capture more of the surplus created by the network.

5 Identification and empirical design

5.1 What identifies bargaining power

Identification comes from variation in which plans contract with which hospitals, conditional on estimated producer surplus. If two hospitals generate similar surplus for plans, but one is excluded less often or appears to require higher surplus to be included, the model infers that the latter hospital has greater bargaining power.

Interpretation: The idea is revealed preference. Observed inclusions and exclusions reveal bounds on hospital markups because a plan accepts a contract only if the expected profit from accepting exceeds the profit from rejecting.

5.2 Why network choices can reveal prices without observing prices

The paper does not observe negotiated prices. Instead, it observes whether a contract exists. Suppose a plan does not contract with a hospital even though the estimated gross surplus of adding the hospital is positive. The model interprets this as evidence that the hospital demanded a payment larger than the gross surplus increment.

Interpretation: Noncontracts are especially informative. Producer surplus alone explains observed contracts reasonably well, but it performs poorly for explaining missing contracts. Adding bargaining terms improves the explanation of noncontracts substantially.

5.3 Role of demand estimates

Demand estimates provide the link between network structure and profit. They determine:

- how much plan demand rises when a hospital is added;
- how much premiums can adjust when networks change;
- how patients shift across hospitals inside the network;
- whether a hospital is a star provider;
- whether a hospital is predicted to be capacity constrained.

Interpretation: The demand model is not a side exercise. It is the engine that converts a counterfactual network into a predicted surplus change.

5.4 Why small deviations are used

The main inequalities compare the observed network to alternatives that reverse one hospital contract at a time. This avoids solving the full dynamic renegotiation process after large network changes.

Interpretation: Small deviations make the revealed-preference restrictions credible and computationally feasible. They also match the intuition that a plan is comparing whether to add or drop particular providers from a network.

5.5 Handling multiple equilibria and endogeneity

The method avoids selecting a single equilibrium. It does not require solving a complete two-sided matching problem. Instead, it uses necessary conditions that must hold in any equilibrium. It also allows measurement error and plan prediction errors, then uses moment inequalities rather than point equations.

Interpretation: This is why the paper reports confidence intervals for identified sets rather than relying only on point estimates. The estimates are informative about the direction and approximate magnitude of bargaining effects, but precise magnitudes are harder to pin down.

5.6 Key assumptions

The main assumptions are:

- plans choose networks to maximize expected profits;
- hospitals and plans have passive beliefs about unobserved contract offers;

- the unobserved profit errors are mean zero conditional on firms' information sets;
- observed hospital and plan characteristics used as instruments are known to firms and observed without relevant error;
- estimated demand and costs are accurate enough to generate meaningful surplus comparisons.

Interpretation: These assumptions are strong but transparent. The paper's robustness checks test whether the qualitative conclusions survive alternative bargaining specifications and alternative hospital-characteristic measures.

5.7 Main threats to interpretation

- **Unobserved physician networks.** Plans may exclude hospitals because contracting with physicians associated with those hospitals is costly.
- **Demand-estimation error.** Star and capacity variables are generated from the demand model, so misspecification can affect their estimated bargaining coefficients.
- **Unobserved plan costs.** Nonhospital costs are not observed and must be modeled or absorbed.
- **Passive beliefs and renegotiation.** Real bargaining may include follow-up offers, whereas the main model simplifies the dynamic negotiation process.
- **Set identification.** The model identifies ranges of plausible parameters rather than exact values.

Interpretation: The robustness checks are important because the empirical design is structural and revealed-preference based, not a randomized experiment.

6 Tables and figures

6.1 Table 1: Descriptive Statistics for Hospitals; Table 2: Descriptive Statistics for HMO/POS Plans

Read:

- Table 1 shows that the average hospital has about 339 beds and 1.26 registered nurses per bed; about 20 percent are teaching hospitals.
- The service variables show that cardiac, imaging, cancer, and birth services vary substantially across hospitals.
- Table 2 shows that average HMO/POS plan market share is about 3 percent, while average premiums are about \$141 per member per month.
- The plan data contain substantial heterogeneity in clinical and satisfaction measures, plan age, and plan type.

Economic message: These tables define the observable heterogeneity used in the demand and supply models. The paper's identification comes from how these characteristics translate into networks and exclusions.

Interpretation: Hospital characteristics matter for patient demand; plan characteristics matter for plan demand. The analysis needs both because bargaining value depends on the interaction between hospitals' attractiveness and plans' ability to turn access into enrollments and premiums.

	Mean	Standard deviation
Number of beds (set up and staffed)	338.66	217.19
Teaching status	0.195	0.397
For-profit	0.202	0.401
Registered nurses per bed	1.263	0.498
Cardiac services	0.812	0.310
Imaging services	0.539	0.287
Cancer services	0.647	0.402
Birth services	0.857	0.348

Notes: $N = 665$ hospitals. Cardiac, imaging, cancer, and birth services refer to four summary variables defined in Table 4. Each hospital is rated on a scale from 0 to 1, where 0 indicates that no procedures in this category are provided by the hospital, and a higher rating indicates that a less common service is offered.

(a) Table 1: Descriptive Statistics for Hospitals.
Original descriptive tables.

Variable	Definition	N	Mean	Standard deviation
Market share	Plan share of nonelderly market	516	0.03	0.04
Premium pmprn (\$)	Premiums earned per member per month	478	140.75	44.27
Physicians per 1,000 population	Number of physician contracts per 1,000 population in markets covered by plan	418	1.56	1.51
Breast cancer screening	Percent of women age 52–69 who received a mammogram within last 2 years	352	0.73	0.05
Cervical cancer screening	Percent of adult women who received pap smear within last 3 years	352	0.72	0.07
Check-ups after delivery	Percent of new mothers receiving a check-up within 8 weeks of delivery	351	0.72	0.11
Eye exams for diabetics	Percent of adult diabetics receiving eye exam within last year	350	0.45	0.11
Adolescent immunization 1	Percent of children receiving all required doses of MMR and Hep B vaccines before 13th birthday	346	0.31	0.16
Adolescent immunization 2	Percent of children receiving all required doses of MMR, Hep B, and VZV vaccines before 13th birthday	313	0.15	0.11
Advice on smoking	Percent of adult smokers advised by physician to quit	213	0.63	0.07
Mental illness checkup	Percent of members seen as outpatient within 30 days of discharge after hospitalization for mental illness	307	0.68	0.15
Care quickly	Composite measure of member satisfaction re: getting care as soon as wanted	304	0.75	0.05
Care needed	Composite measure of member satisfaction re: getting authorizations for needed/desired care	304	0.72	0.06
Age 0–2	Dummy for plans age 0–2 years	516	0.01	0.08
Age 3–5	Dummy for plans age 3–5 years	516	0.06	0.23
Age 6–9	Dummy for plans age 6–9 years	516	0.17	0.37
Aetna	Plan fixed effect	516	0.15	0.36
CIGNA	Plan fixed effect	516	0.10	0.31
Kaiser	Plan fixed effect	516	0.03	0.16
Blue Cross/Blue Shield	Dummy for ownership by BCBS	516	0.16	0.36
POS plan	Dummy for POS plan	516	0.35	0.49

The final dataset analyzed in this paper defines the network of hospitals offered to enrollees

(b) Table 2: Descriptive Statistics for HMO/POS Plans.

6.2 Figure 1: Variation in Plan Networks across and within Markets; Table 3: Summary Data for Selective and Unselective Markets

Read:

- Figure 1 documents substantial variation in how many major hospitals plans exclude.
- Many plans exclude no major hospitals, but a nontrivial set excludes several major providers.
- The second panel shows that selective markets contain more plans excluding multiple major hospitals.
- Table 3 shows that selective and unselective markets look similar on most observables.
- The main significant difference is the standard deviation of distances between hospitals, suggesting that hospital clustering may matter for substitution and exclusion.

Economic message: The figure and table motivate the supply-side analysis: network selectivity varies in ways not fully explained by simple market characteristics.

Interpretation: If selective markets were simply smaller, poorer, more managed-care intensive, or more hospital-rich, network differences might reflect observables. Table 3 reduces that concern and points toward demand substitution and bargaining as explanations.

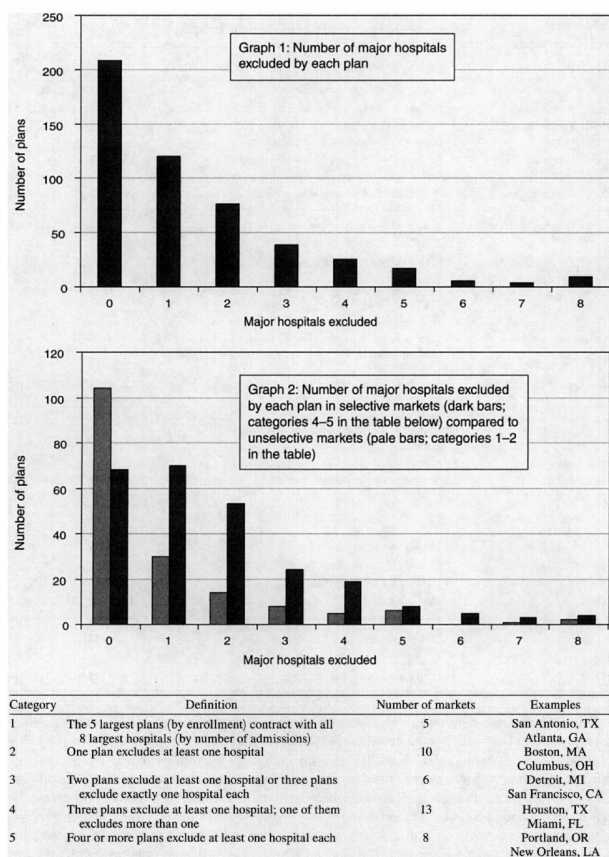


FIGURE 1. VARIATION IN PLAN NETWORKS ACROSS AND WITHIN MARKETS

Note: This figure summarizes the variation in selectivity of plans' hospital networks both across and within markets.

(a) Figure 1: Variation in Plan Networks across and within Markets.

Network variation and market comparison.

TABLE 3—SUMMARY DATA FOR SELECTIVE AND UNSELECTIVE MARKETS

	Unselective markets (category 1 and 2) mean (std dev)	Selective markets (category 4 and 5) mean (std dev)	p-value for difference in means
Market population (million)	2.36 (1.11)	2.36 (1.96)	1.00
Number of HMO/POS plans with over 1 percent market share	6.80 (1.70)	6.57 (1.89)	0.71
Number of hospitals	19.80 (11.40)	21.24 (20.53)	0.78
Beds per 1,000 population	2.78 (1.00)	2.90 (0.99)	0.74
Managed care penetration	0.33 (0.17)	0.35 (0.15)	0.66
Average age of population	34.76 (2.19)	34.31 (1.39)	0.49
Percent of under 65 population age 55–64	0.09 (0.01)	0.09 (0.01)	0.75
Median total family income of population	\$48,890 (\$8,460)	\$46,130 (\$8,642)	0.35
Standard deviation of total family income of population	\$53,687 (\$9,805)	\$52,797 (\$6,511)	0.76
Mean distance between hospitals (miles)	11.71 (5.60)	13.41 (5.12)	0.36
Standard deviation of distances between hospitals (miles)	7.67** (3.37)	10.30** (4.06)	0.04
Number hospitals with open heart surgery	8.07 (3.67)	10.19 (8.59)	0.31
N	15	21	

**Significant difference in means at the 5 percent level

(b) Table 3: Summary Data for Selective and Unselective Markets.

6.3 Table 4: Definition of Hospital Services; Table 5: Relation of Hospital Characteristics to Market Shares

Read:

- Table 4 defines the service indices used to summarize hospital capabilities in cardiac care, imaging, cancer care, and births.
- Table 5 shows that all four service indices and teaching status are positively related to hospital market shares.
- With market fixed effects, the adjusted R^2 rises to 0.69, showing that hospital attractiveness is partly local-market specific.

Economic message: Service capabilities and teaching status are demand-relevant hospital attributes. These are precisely the types of attributes that can create provider bargaining power.

Interpretation: Table 5 validates the demand-side logic: hospitals that offer more sophisticated or broader services have larger market shares. The supply-side model later asks whether such demand appeal turns into higher markups.

Cardiac	Imaging	Cancer	Births
1. Cardiac catheterization lab	1. Ultrasound	1. Oncology services	1. Obstetric care
2. Cardiac intensive care	2. CT scans	2. Radiation therapy	2. Birthing room
3. Angioplasty	3. MRI		
4. Open heart surgery	4. SPECT		
	5. PET		

Notes: This table sets out the definition of the hospital service variables summarized in Table 1. Hospitals were rated on a scale from 0 to 1 within four service categories, where 0 indicates that no services within this category are provided by the hospital, and a higher rating indicates that less common (assumed to be higher-tech) service in the category is offered. The categories are cardiac, imaging, cancer, and births. The services included in each category are listed in Table 5. The exact methodology for rating hospitals is as follows. If the hospital provides none of the services, its rating = 0. If it provides the least common service, its rating = 1. If it offers some service X but not the least common service, its rating = $(1 - x)/(1 - y)$, where x = the percent of hospitals offering service X and y = the percent of hospitals offering the least common service.

(a) Table 4: Definition of Hospital Services.

Service definitions and their demand relevance.

	Coefficient estimate	
Cardiac services	0.732** (0.104)	0.676** (0.072)
Imaging services	0.233** (0.107)	0.224** (0.074)
Cancer services	0.158** (0.079)	0.299** (0.054)
Birth services	0.507** (0.082)	0.394** (0.056)
Teaching hospital	0.243** (0.074)	0.461** (0.051)
Constant	-4.484** (0.097)	-0.005 (0.007)
Market FEs?	No	Yes
Adjusted R^2	0.27	0.69

Notes: Regression of the log of hospital market shares on hospital characteristics. N = 633 hospitals (the 665 providers in the full dataset less 14 Kaiser hospitals and 18 hospitals in Baltimore, MD, which were excluded from the supply-side analysis). Standard errors are reported in parentheses. Cardiac, imaging, cancer, and birth services refer to the four hospital service variables defined in Table 4.

**Significant at the 5 percent level.
*Significant at the 10 percent level.

(b) Table 5: Relation of Hospital Characteristics to Market Shares.

6.4 Table 6: Results of Plan Demand Estimation; Table 7: Results of Full Model for Estimation

Read:

- Table 6 shows that expected utility from the hospital network has a positive and significant effect on plan demand.
- The premium coefficient is negative but imprecise, while several quality measures have significant coefficients.
- Table 7 is the main supply-side result. Star hospitals have higher per-patient markups, and predicted capacity-constrained hospitals also have high estimated markups.
- Hospitals in systems receive a positive fixed component, and dropping a same-system hospital is costly for plans.
- The cost per admission coefficient is negative: higher-cost hospitals receive lower markups, consistent with bargaining models where low-cost providers can capture more surplus.

Economic message: Table 6 establishes why networks matter for plans. Table 7 then shows how this demand value is divided: attractive, scarce, and system hospitals capture more surplus.

Interpretation: Read Table 7 as the paper's central bargaining evidence. The model does not observe prices directly; it infers hospital markups from which contracts are accepted and rejected. The signs line up with the theory that differentiated hospitals and systems have leverage.

TABLE 6—RESULTS OF PLAN DEMAND ESTIMATION

	Coefficient estimate
Premium (\$00 pmpm)	−0.94 (1.13)
Expected utility from hospital network (EU_{rep_m} or $EU_{(m)}$)	0.59** (0.21)
Premium (\$00 pmpm) / Income (\$000 per year)	0.002 (43.9)
Physicians per 1,000 population	0.21** (0.09)
Breast cancer screening	−0.38 (2.66)
Cervical cancer screening	4.40** (2.09)
Check-ups after delivery	0.18 (1.38)
Eye exams for diabetics	−1.19 (1.60)
Adolescent immunization 1	−4.11** (1.17)
Adolescent immunization 2	3.08 (3.76)
Advice on smoking	6.17** (2.08)
Mental illness check-ups	2.70** (1.30)
Care quickly	0.78 (5.63)
Care needed	0.85 (3.99)
Plan age: 0–2 years	1.36 (0.97)
Plan age: 3–5 years	−0.64 (1.97)
Plan age: 6–9 years	−0.25 (0.58)
POS plan	−1.11** (0.13)
Constant	−10.50* (5.65)
Large plan fixed effects	Yes
Market fixed effects	Yes

Notes: $N = 559$ plans (the 516 HMO/POS plans in the full dataset plus one indemnity/PPO option in each market). Standard errors (adjusted for the three-stage estimation process) are reported in parentheses.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

TABLE 7—RESULTS OF FULL MODEL FOR ESTIMATION

	Star hospitals	Predicted cap con	Add number of enrollees	Multinomial logit
Hospital characteristics				
Number of enrollees			0.011	
Simulated 95 percent C.I.			(−0.06, 0.09)	
Conservative 95 percent C.I.			(−0.17, 0.11)	
Fixed component (unit = \$ million per month)				
Hospital in system	0.18	0.16	0.17	−0.60
Simulated 95 percent C.I.	(0.05, 0.37)	(0.03, 0.34)	(0.02, 0.34)	(0.09)
Conservative 95 percent C.I.	(0.18, 0.91)	(0.15, 0.75)	(0.15, 0.66)	
Drop same system hosp	0.09	0.08	0.08	0.62
Simulated 95 percent C.I.	(0.01, 0.16)	(0.01, 0.14)	(0.00, 0.15)	(0.06)
Conservative 95 percent C.I.	(0.04, 0.35)	(0.04, 0.28)	(0.06, 0.29)	
Per patient component (unit = \$ thousand per patient)				
Constant	5.51	14.90	6.08	−1.41
Simulated 95 percent C.I.	(−1.93, 16.6)	(1.03, 19.3)	(−2.70, 13.2)	(3.29)
Conservative 95 percent C.I.	(−24.0, 21.5)	(−22.7, 27.9)	(−12.8, 31.4)	
Star hospital	6.77		4.69	7.73
Simulated 95 percent C.I.	(3.27, 15.8)		(0.78, 16.4)	(1.41)
Conservative 95 percent C.I.	(6.40, 27.0)		(4.68, 33.3)	
Cost per admission	−0.80	−1.47	−0.74	−0.39
Simulated 95 percent C.I.	(−1.96, −0.32)	(−2.02, −0.41)	(−1.77, −0.31)	(0.19)
Conservative 95 percent C.I.	(−3.40, 0.16)	(−3.55, −0.32)	(−3.09, 0.04)	
Capacity constrained		6.92		
Simulated 95 percent C.I.		(3.64, 15.7)		
Conservative 95 percent C.I.		(5.36, 24.4)		

Notes: $N = 441$ insurance plans (the 516 in the full dataset less 9 plans in Baltimore, MD; 13 Kaiser plans; 42 with unobserved premiums; and 8 selective plans that I regard as outliers). Figures in parentheses represent 95 percent confidence intervals (C.I.). Plan premium adjustments are incorporated as described in Section IID. Coefficients represent the predicted profits to the hospital. “Drop same system hosp” is an indicator for hospitals for which a same-system hospital has been excluded. Star hospitals have above ninetieth percentile market share when all plans contract with all hospitals; capacity-constrained hospitals are full under the same thought experiment. “Number of enrollees” is the model’s prediction of the plan’s number of enrollees given its premium and network choice. The logit results report standard errors calculated using the usual maximum likelihood estimation methodology.

(a) Table 6: Results of Plan Demand Estimation.

Demand estimation and main bargaining estimates.

(b) Table 7: Results of Full Model for Estimation.

6.5 Table 8: Adding Plan and Market Characteristics; Table 9: Alternative Specifications

Read:

- Table 8 adds market and plan characteristics to the hospital profit equation.
- Star hospital effects remain positive and significant across specifications.
- Market beds per population has a negative coefficient, consistent with the idea that hospital scarcity increases bargaining leverage.
- Table 9 shows that the qualitative conclusions survive alternative assumptions about premium adjustment, plan/hospital response, and objective functions for nonprofit/public entities.

Economic message: These robustness exercises show that the main results are not artifacts of a single specification. Market scarcity and provider differentiation remain important.

Interpretation: Table 8 helps separate star appeal from market-level scarcity. Table 9 checks the bargaining environment. Together they show that the star/system/cost patterns are robust enough to support the paper’s qualitative conclusions.

Hospital characteristics	Beds per population	Number of patients	Local plan	Plan: good screening
<i>Per patient component (unit = \$ thousand per patient)</i>				
Constant	8.48	6.26	5.60	7.33
Simulated 95 percent C.I.	(−3.95, 31.9)	(−3.64, 15.9)	(−2.43, 15.5)	(−1.18, 13.6)
Conservative 95 percent C.I.	(−8.05, 58.4)	(−6.86, 20.9)	(−5.78, 21.5)	(−8.16, 21.8)
Star hospital	8.48	6.93	6.36	8.56
Simulated 95 percent C.I.	(1.97, 16.1)	(2.40, 14.9)	(2.12, 12.6)	(2.69, 11.7)
Conservative 95 percent C.I.	(7.36, 35.2)	(5.69, 19.7)	(5.59, 18.9)	(7.67, 23.1)
Cost per admission	−0.76	−0.76	−0.74	−0.93
Simulated 95 percent C.I.	(−2.56, −0.02)	(−1.69, 0.02)	(−1.85, −0.23)	(−1.65, −0.24)
Conservative 95 percent C.I.	(−4.36, 0.52)	(−2.70, −0.16)	(−2.96, −0.24)	(−3.0, −0.27)
Market beds per population	−1.19			
Simulated 95 percent C.I.	(−5.14, 1.70)			
Conservative 95 percent C.I.	(−15.3, −1.17)			
Number of patients		−0.54		
Simulated 95 percent C.I.		(−4.66, 11.0)		
Conservative 95 percent C.I.		(−3.31, 16.2)		
Local plan			1.21	
Simulated 95 percent C.I.			(−0.37, 11.4)	
Conservative 95 percent C.I.			(−7.06, 15.2)	
Breast cancer screen				135.7
Simulated 95 percent C.I.				(−79.9, 251)
Conservative 95 percent C.I.				(−104, 439)

Notes: Results from including plan and market characteristics in per patient profits. $N = 441$ insurance plans. “Beds per population” is the number of hospital beds in the market per 1,000 population. “Number of patients” is the num-

(a) Table 8: Adding Plan and Market Characteristics.

Hospital characteristics	No prem adjustments	Allow plans to adjust	Allow hosps to adjust	NFP plan obj. finance	Public hosp obj. finance
<i>Fixed component (unit = \$ million per month)</i>					
Hospital in system	0.20	0.07	0.17	0.17	0.12
Simulated 95 percent C.I.	(0.05, 0.41)	(0.02, 0.32)	(0.04, 0.35)	(0.04, 0.33)	(0.04, 0.30)
Conservative 95 percent C.I.	(0.20, 1.03)	(0.07, 0.62)	(0.17, 0.75)	(0.14, 0.82)	(0.12, 0.64)
Drop same system hospital	0.09	0.04		0.08	0.07
Simulated 95 percent C.I.	(0.01, 0.18)	(0.01, 0.15)	(0.01, 0.15)	(0.01, 0.15)	(0.02, 0.13)
Conservative 95 percent C.I.	(0.04, 0.41)	(0.01, 0.26)	(0.05, 0.32)	(0.02, 0.31)	(0.04, 0.27)
<i>Per patient component (unit = \$ thousand per patient)</i>					
Constant	9.65	9.26	8.15	5.58	9.23
Simulated 95 percent C.I.	(−1.73, 16.7)	(−0.69, 14.5)	(1.41, 18.0)	(−4.17, 17.1)	(−3.62, 16.4)
Conservative 95 percent C.I.	(−20.6, 15.3)	(−14.2, 17.0)	(−14.3, 19.1)	(−22.0, 22.7)	(−21.4, 23.0)
Star hospital	2.76	2.21	2.28	6.64	4.14
Simulated 95 percent C.I.	(1.37, 13.1)	(1.17, 10.4)	(1.39, 13.3)	(0.95, 12.7)	(2.83, 11.8)
Conservative 95 percent C.I.	(2.68, 23.1)	(2.19, 15.6)	(2.19, 20.6)	(6.34, 24.3)	(4.04, 20.7)
Cost per admission	−0.99	−0.90	−0.85	−0.84	−0.85
Simulated 95 percent C.I.	(−1.94, −0.42)	(−1.65, −0.26)	(−1.87, −0.48)	(−1.7, −0.23)	(−1.56, −0.17)
Conservative 95 percent C.I.	(−2.74, −0.07)	(−2.28, 0.04)	(−2.5, −0.18)	(−3.1, −0.05)	(−2.59, 0.07)
NFP plan × enrollees				0.31	
Simulated 95 percent C.I.				(−2.24, 3.15)	
Conservative 95 percent C.I.				(−3.23, 6.13)	
Public hospital × patients					12.38
Simulated 95 percent C.I.					(−19.4, 542)
Conservative 95 percent C.I.					(−30.7, 523)

Notes: Results with alternative specifications. $N = 441$ insurance plans. Figures in parentheses represent 95 percent confidence intervals (C.I.). Column 1 gives the results when plans do not adjust premiums in response to network changes. Columns 2 and 3 allow plans and hospitals, respectively, to respond by changing networks after a deviation by their negotiating partners. Plans are not permitted to adjust premiums in these scenarios, so the results should be compared to column 1. See Sections VA and VD for details. Columns 4 and 5 allow not-for-profit plans and public hospitals to maximize a weighted sum of profits and the number of enrollees/number of patients treated, respectively.

(b) Table 9: Alternative Specifications.

Robustness to market/plan controls and alternative bargaining assumptions.

6.6 Table 10: Robustness to Choice of Hospital Characteristics; Table 11: Investigating Fit of the Model

Read:

- Table 10 tests whether the star and capacity results are sensitive to alternative definitions, including PET imaging and lagged capacity constraints.
- The results remain qualitatively consistent, though some coefficients become smaller and less precise.
- Table 11 compares fit using producer surplus alone versus the full bargaining model.
- Producer surplus alone explains 70.2 percent of observed contracts but only 35.8 percent of missing contracts.
- The full model explains 72.5 percent of observed contracts and 54.0 percent of missing contracts.

Economic message: The full model adds explanatory power mainly for explaining why contracts are not formed. That is exactly where bargaining should matter most.

Interpretation: Table 11 is an important validation exercise. If producer surplus alone were enough, bargaining terms would be unnecessary. The large improvement for noncontracts shows that hospital profits and bargaining power are central to network formation.

TABLE 10—ROBUSTNESS TO CHOICE OF HOSPITAL CHARACTERISTICS

Hospital characteristics	Main specification	Star and cap con	Imaging services	Cap con last year
<i>Fixed component (unit = \$ million per month)</i>				
Hospital in system	0.18	0.12	0.33	0.39
Simulated 95 percent C.I.	(0.05, 0.37)	(0.04, 0.35)	(0.04, 0.54)	(0.06, 0.62)
Conservative 95 percent C.I.	(0.18, 0.91)	(0.12, 0.64)	(0.28, 1.13)	(0.31, 1.4)
Drop same system hospital	0.09	0.06	0.15	0.17
Simulated 95 percent C.I.	(0.01, 0.16)	(0.01, 0.15)	(0.03, 0.24)	(0.03, 0.25)
Conservative 95 percent C.I.	(0.04, 0.35)	(0.02, 0.24)	(0.10, 0.48)	(0.12, 0.57)
<i>Per patient component (unit = \$ thousand per patient)</i>				
Constant	5.51	9.09	3.14	1.37
Simulated 95 percent C.I.	(−1.93, 16.6)	(−5.36, 15.8)	(−9.77, 17.0)	(−13.6, 17.1)
Conservative 95 percent C.I.	(−24.0, 21.5)	(−17.4, 17.2)	(−25.6, 19.3)	(−53.4, 18.8)
Star hospital	6.77	3.48		
Simulated 95 percent C.I.	(3.27, 15.8)	(−0.72, 14.0)		
Conservative 95 percent C.I.	(6.40, 27.0)	(0.35, 27.0)		
Cost per admission	−0.80	−1.10	−0.38	−0.35
Simulated 95 percent C.I.	(−1.96, −0.32)	(−1.89, −0.28)	(−1.56, 0.27)	(−1.63, 0.70)
Conservative 95 percent C.I.	(−3.40, 0.16)	(−2.76, −0.16)	(−2.23, 0.97)	(−2.41, 2.10)
Capacity constrained		4.23		
Simulated 95 percent C.I.		(−6.21, 12.6)		
Conservative 95 percent C.I.		(−12.9, 16.7)		
Imaging services		2.85		
Simulated 95 percent C.I.		(−2.51, 7.90)		
Conservative 95 percent C.I.		(−4.30, 14.4)		
Last year cap con			1.72	
Simulated 95 percent C.I.			(−3.20, 8.07)	
Conservative 95 percent C.I.			(−7.09, 21.4)	

Notes: Results of robustness tests. $N = 441$ insurance plans. The first test includes both star and capacity-constrained hospitals. The second replaces the star variable with an indicator for the 22 percent of hospitals that offer positron emission tomography. The third replaces predicted capacity constraints with an indicator for capacity constraints in the previous year.

TABLE 11—INVESTIGATING FIT OF THE MODEL

	Contract observed	Contract not agreed upon
Number of contracts	5,587	1,160
Percent explained by $\Delta surplus$	70.2 percent	35.8 percent
Percent explained by full model	72.5 percent	54.0 percent

Notes: Summary of ability of the model to explain the data. “Percent explained by $\Delta surplus$ ” is the percent of observed contracts for which $\Delta surplus > 0$ (i.e., for which the producer surplus generated by the contract is positive) and the percent of contracts not agreed upon for which $\Delta surplus < 0$. “Percent explained by full model” has analogous meaning, with $\Delta surplus$ replaced by $\Delta \pi$, the profit change predicted by the full model.

(a) Table 10: Robustness to Choice of Hospital Characteristics. (b) Table 11: Investigating Fit of the Model.

Robustness of hospital-characteristic measures and model fit.

7 Short summary

- Ho studies insurer-provider network formation in managed care markets.
- The paper combines hospital-choice estimates, plan-choice estimates, hospital cost data, and observed plan-hospital contracts.
- The main empirical innovation is to infer hospital markups without observing negotiated prices.
- The method uses revealed-preference inequalities from observed networks.
- Producer surplus is calculated for counterfactual hospital networks using demand estimates.
- Positive producer surplus is not sufficient for a contract if hospitals demand too much of the surplus.
- Star hospitals, capacity-constrained hospitals, and hospital systems have higher bargaining leverage.
- Hospitals in systems also appear to penalize plans that contract with only some system members.
- Higher-cost hospitals receive lower markups.
- The full bargaining model explains missing contracts much better than producer surplus alone.
- The results imply that hospitals have incentives to invest in demand-enhancing services, join systems, and perhaps strategically limit capacity.

8 Conclusion

8.1 Core conclusion

The paper shows that insurer-provider networks are shaped by bargaining, not only by the gross surplus created by hospital access. Plans value broad and attractive hospital networks because networks affect consumer demand, but high-value hospitals can capture a large share of the surplus they create. This explains why some positive-surplus contracts may not be observed.

8.2 Economic intuition

The basic intuition is that hospital access is both a product attribute and a bargaining asset. A plan can attract consumers by including a hospital, but if the hospital is highly valued, scarce, or part of a system, the hospital can demand a higher payment. The plan may then choose exclusion even when consumers would value inclusion.

8.3 Incremental lesson for healthcare markets

The paper moves the analysis of managed care beyond the question of whether plans reduce medical spending. It shows that network design itself is an equilibrium outcome of vertical bargaining. Hospital investment, hospital mergers, and capacity choices can all change bargaining leverage, so the long-run consequences of network contracting depend on hospitals' strategic responses.

8.4 Policy implications

The results imply that hospital mergers can raise bargaining power even if they do not obviously reduce patient choice in a simple static sense. The results also suggest that network adequacy rules, any-willing-provider laws, and antitrust policy should consider bargaining incentives, not only consumer access. If systems can force plans to contract with all system members, plans may lose an important tool for limiting provider prices.

8.5 Limitations

The paper does not observe negotiated prices, physician networks, or the full bargaining protocol. It uses estimated demand as an input, and the supply-side estimates are set identified. Therefore, the results are strongest as evidence about the direction and relative importance of bargaining forces, rather than as exact measurements of every contract price.

8.6 Takeaway

Hospital networks are a revealed outcome of demand, costs, and bargaining power. Ho's framework shows how to use those network choices to learn about the division of surplus in a vertical healthcare market. The main takeaway is that attractive hospitals, scarce hospitals, and hospital systems can capture substantial markups, and these markups shape both short-run network design and long-run hospital incentives.